

Zachary Yedidia

Research interests: operating systems, security, compilers, computer architecture.

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EDUCATION

Stanford University

Ph.D. candidate in Computer Science
Advisor: Prof. David Mazières

Stanford, CA
Sep. 2021 – June 2027 (expected)

Harvard University

S.M. in Computer Science

Cambridge, MA
Sep. 2020 – May 2021

Harvard College

A.B. Magna cum laude with highest honors in Computer Science

Cambridge, MA
Aug. 2017 – May 2021

PUBLICATIONS

LFI-XL: Scaling Software-Based Sandboxing to Modern Applications

Zachary Yedidia, Abhishek Sharma, Anand Balaji, Taehyun Noh, Tal Garfinkel, Hovav Shacham, Shravan Narayan

IEEE S&P 2027
To appear

Mohabi: Disaggregating and Sandboxing the Firefox JavaScript Engine

Abhishek Sharma, Anand Balaji, Zachary Yedidia, Anthony Du, Taehyun Noh, Iain Ireland, Jan de Mooij, Matthew Gaudet, Tal Garfinkel, Deian Stefan, Hovav Shacham, Shravan Narayan
Honorable mention for best paper award

OSDI 2026

Automated Formal Verification of a Software Fault Isolation System (short paper)

Matthew Sotoudeh and Zachary Yedidia

FMCAD 2025

Deterministic Client: Enforcing Determinism on Untrusted Machine Code

Zachary Yedidia, Geoffrey Ramseier, David Mazières

OSDI 2025

Segue & ColorGuard: Optimizing SFI Performance and Scalability on Modern Architectures

Shravan Narayan, Tal Garfinkel, Evan Johnson, Zachary Yedidia, Yingchen Wang, Andrew Brown, Anjo Vahldiek-Oberwagner, Michael LeMay, Wenyong Huang, Xin Wang, Mingqiu Sun, Dean Tullsen, Deian Stefan

ASPLOS 2025

DeCl: Deterministic and Metered Native Sandboxes

Zachary Yedidia, Geoffrey Ramseier, David Mazières

SIB 2024
Workshop

Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing

Zachary Yedidia

ASPLOS 2024

Fast Incremental PEG Parsing

Zachary Yedidia and Stephen Chong
Best paper award

SLE 2021

Precision Batching: Bitserial Decomposition for Efficient Neural Network Inference on GPUs

Maximilian Lam, Zachary Yedidia, Colby Banbury, Vijay Janapa Reddi

PACT 2021

OPEN SOURCE PROJECTS

Lightweight Fault Isolation (LFI)

Software-based sandboxing system that enforces memory isolation on machine code within a single address space. Used for in-process library isolation, serverless functions, and smart contracts. Upstreamed into LLVM and used in Android.

[Paper](#), [Code](#)

LLVM

Maintainer of the [LFI](#) feature in LLVM, including the AArch64 and x86-64 LFI targets. 29 commits with 8,500+ lines written.

[Docs](#), [Code](#)

Micro Text Editor

[Website](#), [Code](#)

Terminal-based text editor I created in 2016 as a modern successor to Nano. It has 25,000+ GitHub stars, 1,000,000+ downloads, and 300+ contributors. Micro is available in Homebrew, Apt, Arch Linux, Snap, and more, and is the default editor on Garuda Linux. Also wrote [Mu](#), a personal rewrite tailored to my preferences that I use daily.

Arcilator · A hardware simulator that I helped develop as part of the CIRCT project while at SiFive.

[Code](#)

GPeg · Incremental PEG parsing library from my senior thesis with Prof. Stephen Chong.

[Paper](#), [Thesis](#), [Slides](#), [Code](#)

Literate · A tool for compiling Literate programs written in any programming language.

[Website](#), [Code](#)

Eget · A tool for automatically installing pre-built binaries distributed in GitHub releases.

[Code](#)

Go Generic Data Structures · A library of generic data structures for Go.

[Code](#)

Knit · A flexible build tool that combines Lua with Make's declarative rules language.

[Code](#)

Multiplix · A small operating system for AArch64 and RISC-V SBCs written in D.

[Code](#)

Riscinator · A simple 3-stage pipelined RISC-V core and SoC implemented in Chisel.

[Code](#)

SystemVerilog Guide · Course materials for CS 141 (Spring 2020) at Harvard.

[PDF](#)

EXPERIENCE

Google · Student Researcher (Deploying LFI in Android)

Mountain View, CA · *Summer 2025 – Present*

SiFive · Compiler Intern ([Chisel](#) and [CIRCT](#))

San Mateo, CA · *Summer 2022*

Zero ASIC · Intern ([Silicon Compiler](#))

Cambridge, MA · *Summer 2021*

Harvard University · HCRP Research Fellow (Advisor: Prof. Stratos Idreos)

Cambridge, MA · *Summer 2020*

Princeton University · Research Assistant (Advisor: Prof. Naveen Verma)

Princeton, NJ · *Summer 2019*

Harvard University · PRISE Research Fellow (Advisor: Prof. Eddie Kohler)

Cambridge, MA · *Summer 2018*

HONORS

Honorable mention for best paper award, OSDI 2026 (Mohabi).

NSF Graduate Research Fellowship (2022–2024).

Best paper award, SLE 2021 (Fast Incremental PEG Parsing).

Derek Bok Center teaching award, Harvard CS61 (2020).

Derek Bok Center teaching award, Harvard CS141 (2019).

SERVICE

Organizer of Stanford's weekly TGIF CS Department social event (Autumn 2024 — Present).

External reviewer for ACM TOPLAS (2021).

TEACHING

Operating Systems (CS112/CS212) · CA for Prof. David Mazières

Stanford · *Spring 2026*

Distributed Systems (CS244b) · CA for Prof. David Mazières

Stanford · *Spring 2024*

Advanced Topics in Operating Systems (CS240) · CA for Profs David Mazières and Dawson Engler

Stanford · *Spring 2023*

Using Bits to Control Atoms (CS49n) · CA for Prof. Dawson Engler

Stanford · *Autumn 2021*

Systems Programming and Machine Organization (CS61) · TF for Profs Eddie Kohler and Minlan Yu

Harvard · *Fall 2020*

Computing Hardware (CS141) · TF for Profs David Brooks and Vijay Reddi

Harvard · *Spring 2019, Spring 2020*

Compilers (CS153) · TF for Prof. Stephen Chong

Harvard · *Fall 2019*

TALKS

LFI-XL: Scaling Software Fault Isolation Beyond 4GiB · Bay Area Security Day

Oct. '26

Lightweight Fault Isolation in LLVM, One Year Later · LLVM Developers' Meeting (Quick Talk)

Oct. '26

Lightweight Fault Isolation in LLVM: Current Support and Future Directions · LLVM Memory Safety Workshop

Oct. '26

Kage: device driver isolation in the Linux kernel with LFI · Linux Plumbers Conference (Toolchains Track)

Oct. '26

Securing Android's Audio Codecs with Lightweight Fault Isolation · Google LLVM Summit

Aug. '26

Mohabi: Disaggregating and Sandboxing the Firefox JavaScript Engine · ORCA Tech Talk (with Sharma, Balaji)

Aug. '26

Enabling Bare-Metal Smart Contracts With Deterministic Client · Stanford IORH Mini-Conference

July '26

Quark: Compiler Extensions for Post-Link Instrumentation · Stanford Security Workshop	Apr. '26
Enabling Bare-Metal Smart Contracts With Deterministic Client · Stanford FDCI Seminar	Mar. '26
Lightweight Fault Isolation: From Research to Production Sandboxing · Stanford Security Lunch	Mar. '26
Lightweight Fault Isolation: LLVM Support for Efficient Native Code Sandboxing · LLVM Developers' Meeting	Oct. '25
Lightweight Fault Isolation: LLVM Support for Efficient Native Code Sandboxing · LLVM Memory Safety WG	Sep. '25
Securing Native Libraries on Android with Lightweight Fault Isolation (LFI) · Google LLVM Summit	Aug. '25
Deterministic Client: Enforcing Determinism on Untrusted Machine Code · OSDI	July '25
Deterministic Client: Enforcing Determinism on Untrusted Machine Code · MIT DCI	May '25
Securing Native Libraries on Android with LFI · Qualcomm Product Security Summit	May '25
Deterministic Client: Enforcing Determinism on Untrusted Machine Code · Stanford Security Workshop	Apr. '25
Lightweight Fault Isolation · Software Compartmentalization Community Tech Talk	Apr. '25
Deterministic Client: Enforcing Determinism on Untrusted Machine Code · Stanford Security Lunch	Mar. '25
Sandboxing Native Libraries on Android with Lightweight Fault Isolation (LFI) · Google Android Bootcamp	Feb. '25
DeCl: Deterministic and Metered Native Sandboxes · SIB Workshop	Sep. '24
DeCl: Deterministic and Metered Native Sandboxes · Stanford Software Lunch	July '24
Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing · ASPLOS	Apr. '24
Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing · Stanford Security Workshop	Apr. '24
Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing · Stanford Security Lunch	Nov. '23
Classic Software Fault Isolation and WebAssembly · WebAssembly Research Day	Oct. '23
Multiplex: using D for kernel development · DConf	Sep. '23
Fast Incremental PEG Parsing · SLE	Nov. '21
Fast Incremental PEG Parsing · Stanford Software Lunch	Nov. '21

POSTERS

Beyond Bounds Checks: Reducing Overheads in Embedded WebAssembly Nicole Mulvey, Leon Schuermann, <i>Zachary Yedidia</i> , Philip Levis	OSDI 2026
Lightweight Fault Isolation for RISC-V Nicole Mulvey and <i>Zachary Yedidia</i>	RISC-V Summit 2025
Link-time Instrumentation with Quark Michael Paper and <i>Zachary Yedidia</i>	OSDI 2025
Stateless SFI: Safer and Faster Machine Code Validation <i>Zachary Yedidia</i> , Matthew Sotoudeh, David Mazières	OSDI 2024
Lightweight Fault Isolation: Practical, Efficient, and Secure Software Sandboxing <i>Zachary Yedidia</i>	ASPLOS 2024
Using Debug Hardware to Build Tiny and Effective Dynamic Kernel Checkers <i>Zachary Yedidia</i> , Akshay Srivatsan, Manya Bansal, Dawson Engler	SOSP 2023